

Equivariant Systems Theory and Observer Design for Autonomous Systems

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Outline

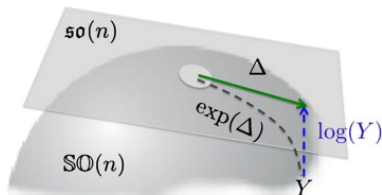
- 1 Velocity
 - Velocity Kinematics on a Lie group
 - Velocity Measurements
- 2 Group Affine Systems and Lifted Groups
 - Group affine definitions
 - Linear Systems
 - Dual Invariant Systems
 - The role of automorphisms
- 3 Note on Observer Design for Group Affine Systems
 - Observer architecture
 - Synchronous Models
- 4 A Taxonomy of Systems

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Matrix Lie-algebra

Think of a **matrix** Lie-group $\mathbf{G}$ as an embedded manifold in matrix $\mathbb{R}^{n \times n}$ space.



Let $X(t) \in \mathbf{G}$ with $X(0) = I$.
Then

$$\left. \frac{d}{dt} \right|_{t=0} X(t) = U \in \mathfrak{g}$$

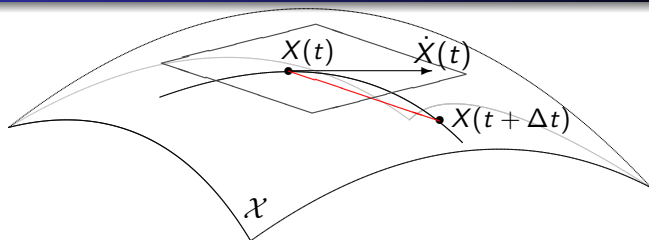
where $\mathfrak{g}$ is the tangent space at the identity.

Here $\mathfrak{g}$ is just a linear subspace of matrix space.

For $\text{SO}(3)$ then the Lie algebra

$$\mathfrak{so}(3) = \{\Omega \in \mathbb{R}^{3 \times 3} \mid \Omega^\top = -\Omega\}$$

Tangent bundle of Lie group



Let $X(t + \tau) \in \mathbf{G}$. Then $X(t)^{-1}X(t + \tau)$ for τ varying is a curve through the identity and

$$\left. \frac{d}{d\tau} \right|_{\tau=0} X(t)^{-1}X(t + \tau) = U(t) \in \mathfrak{g}$$

$$\Updownarrow$$

$$X(t)^{-1}\dot{X}(t) = U(t)$$

Kinematics on Lie group

Differentiating $X(t)^{-1}X(t + \tau)$ with respect to τ at zero yields

$$\dot{X}(t) = X(t)U(t) \in T_{X(t)}\mathbf{G}$$

The same construction using $X(t + \tau)X(t)^{-1}$ shows that

$$\dot{X}(t) = W(t)X(t) \in T_{X(t)}\mathbf{G}$$

Both $\dot{X} = XU$ and $\dot{X} = WX$ are termed *first order* or *velocity kinematics* on a Lie-group.

Invariant systems

A system is termed **left invariant** if it can be written

$$\dot{X} = XU$$

for $U := U(v) \in \mathfrak{g}$ for velocity measurements v .

Example: mobile robot.

A system is termed **right invariant** if it can be written

$$\dot{X} = WX$$

for $W := W(v) \in \mathfrak{g}$ for velocity measurements v .

Example: Target tracking with doppler radar.

A system is termed **dual invariant** if it can be written

$$\dot{X} = XU + WX$$

for $W := W(v) \in \mathfrak{g}$ for velocity measurements v .

Example: INS

Measuring velocity

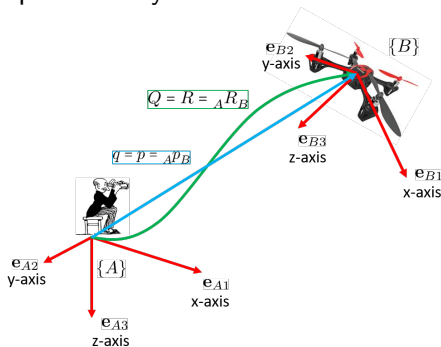
Velocities come from velocity measurements such as doppler radar, doppler acoustic (DVL, inertial measurement units (rate-gyros, accelerometers), optical flow, dynamic pressure (pitot), carrier frequency GPS, etc.

Such systems are themselves physical measurement devices and understanding them as group velocities requires some insight.

How does the physical nature of a velocity measurement device correspond to the mathematical structure of the body and spatial group velocities U and W ?

Rigid body pose

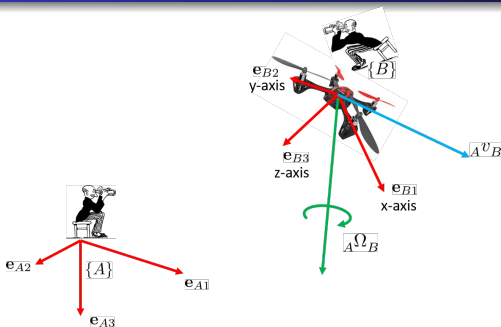
Since rigid-body pose covers almost all the cases of interest and is the most intuitive we will work in this framework to demonstrate the results in a special case. However, the principles apply to all equivariant systems.



Recall the pose of a rigid-body

$$P = \begin{pmatrix} {}^A R_B & {}^A p_B \\ 0 & 1 \end{pmatrix}$$

Velocity



The physical velocities associated with pose are

$$v = {}_A v_B, \quad \Omega = {}_A \Omega_B$$

that is the velocity of target frame $\{B\}$ with respect to reference frame $\{A\}$

We will write the combined pose velocity as

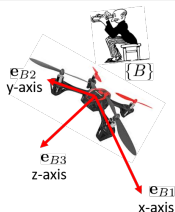
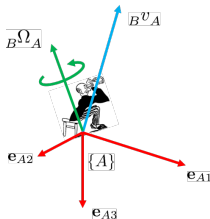
$${}_A U_B = ({}_A \Omega_B, {}_A v_B)$$

Note that velocity of $\{B\}$ with respect to $\{A\}$ is a relative concept and says nothing about absolute velocity of $\{A\}$ or $\{B\}$.

Modelling assumption: Model $\{B\}$ moving relative to $\{A\}$.

What we do know is where the velocity sensor is located!

This matters... a lot



If the velocity sensor is attached to the target/body frame, this is a *body* velocity.

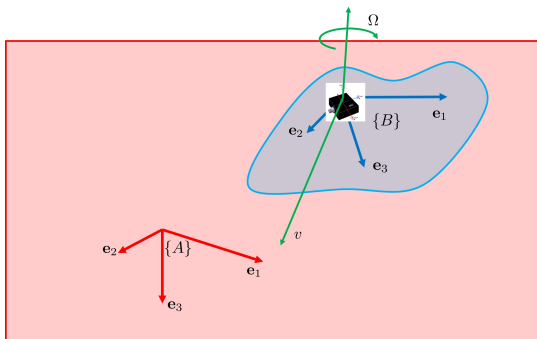
$${}_A U_B^b = ({}_A \Omega_B^b, {}_A v_B^b)$$

If the velocity sensor is attached to the reference frame, this is a *spatial* velocity.

$${}_A U_B^s = ({}_A \Omega_B^s, {}_A v_B^s)$$

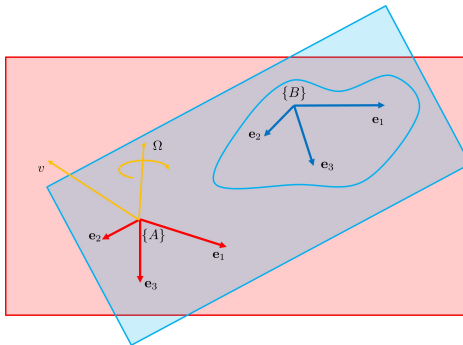
Both velocities encode the same physical relative motion.

Body velocity



A body velocity sensor attached to a target frame (blue) measures its motion (green) relative to a continuum (red) attached to the reference frame (red). The measurement is made at the location of the sensor.

Spatial Velocity



A spatial velocity sensor attached to a reference frame (red) measures the motion (yellow) of a continuum (blue) attached to the target frame (blue). The measurement is made at the location of the sensor.

Velocity Sensors

A body velocity sensor models the world as though it is attached to a moving frame. It measures its motion, at the point it is located, relative to a continuum attached to a static reference frame.

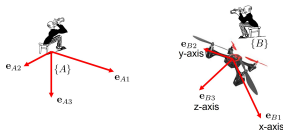
A spatial velocity sensor models the world as though it is stationary and observing a moving world. It measures the motion of the continuum attached to the target frame, at the point it is located.

A body velocity is actually like a spatial velocity where the reference frame is attached to the moving object and the world is moving. However, a **body** velocity sensor outputs the negative of a **spatial** velocity sensor that is colocated. This difference is because of the model assumption for the body velocity sensor that the world is stationary and hence the motion of the body is the negative of the perceived motion of the world.

Sensors

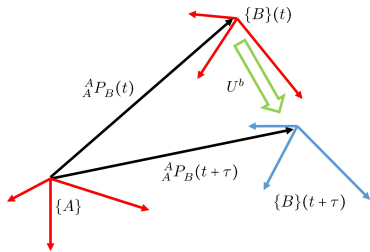
'Velocity' Sensors

Sensor	Symbol	Measurement
Strap down rate gyroscope	Ω	3-axis angular velocity
Strap down accelerometer	a	3-axis linear acceleration
Pitot tube transducer	v_x	airspeed
Optical Flow	ϕ	Scaled velocity
GPS velocity	v	Velocity
Ground based doppler radar	v_r	Range velocity



Excercise: For each sensor; what is type of velocity?

Understanding body velocity measurement



Ignoring the group structure and thinking of AP_B as a matrix

$$U^b = {}^AP_B^{-1} \lim_{\tau \rightarrow 0} \left({}^AP_B(t + \tau) - {}^AP_B(t) \right)$$

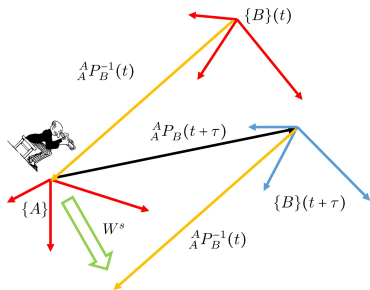
Thus,

$$U^b = {}^AP_B^{-1} {}^A\dot{P}_B$$

Body-fixed velocity measurement devices generate body velocities

$$\dot{P} = P U^b$$

Understanding Spatial velocity measurement



$$W^s = \lim_{\tau \rightarrow 0} \left({}^A P_B(t + \tau) {}^A P_B^{-1}(t) - I \right)$$

Computing this one obtains

$$W^s = {}^A \dot{P}_{BA} {}^A P_B^{-1}$$

Spatial-fixed velocity measurement devices generate Spatial velocities

$$\dot{P} = W^s P$$

Velocity measurement

Typically a body velocity will be measured by a sensor mounted on the autonomous system and will give measurements in the body-frame coordinates.

Typically a spatial velocity will be measured by a sensor mounted on the reference frame and will give measurements in the spatial-frame coordinates.

Both types of sensor will usually require extrinsic (and intrinsic) calibration.

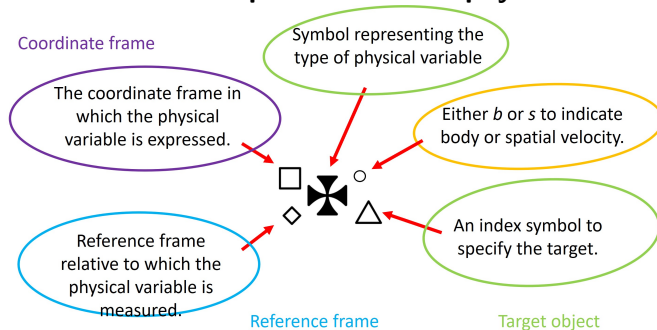
In robotics applications both types of velocities will be seen.

- Strap down gyro measures ${}^B_A\Omega_B^b$
- Doppler GPS measures ${}^A_A v_B^s$

Both velocities refer to the same motion. That is the velocity of $\{B\}$ with respect to $\{A\}$.

Notation

The coordinate expression for the physical velocities



In practice, we mostly use variables without superscripts or subscripts and rely on context for the frame of reference information.

Adjoint

A body velocity measurement can be converted into a spatial velocity measurement by the adjoint operation

$$W^s = XU^bX^{-1} = \text{Ad}_X U^b$$

Since

$$\dot{X} = XU^b = XU^bX^{-1}X = (\text{Ad}_X U^b)X = W^sX$$

The Adjoint operator $\text{Ad}_X : \mathfrak{g} \rightarrow \mathfrak{g}$ converts body velocities to spatial velocities and vice versa.

$${}_A W_B^s = \text{Ad}_{{}_A X_B} {}_A U_B^b = {}_A X_B \left({}_B U_B^b \right) {}_A X_B^{-1}$$

Note: This is not just a change of coordinates.

Velocity Calibration

Consider an autonomous system represented by a frame $\{B\}$ moving with respect to a reference $\{A\}$.

Imagine that you have a velocity sensor in a third frame $\{C\}$, rigidly attached to frame $\{B\}$, that measures the body-velocity

$$V = {}^C_A V_C^b$$

We know that extrinsic calibration transformation ${}^B X_C$ that relates $\{C\}$ to $\{B\}$ with respect to $\{B\}$.

$$\begin{aligned} U &= {}^B_A U_B^b = \text{Ad}_{{}^A X_B^{-1}} {}^A W_B^s = \text{Ad}_{{}^A X_B^{-1}} {}^A W_C^s \\ &= \text{Ad}_{{}^A X_B^{-1}} \text{Ad}_{{}^A X_C} {}^C V_C^b = \text{Ad}_{{}^A X_B^{-1} {}^A X_C} {}^C V_C^b = \text{Ad}_{{}^B X_C} {}^C V_C^b \end{aligned}$$

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Group Affine System

Consider a system

$$\dot{X} = F_v(X)$$

on a Lie-group for a set of inputs (velocity measurements) v .

A system is group-affine if (Bonnabel *et al.*)

$$F_v(XY) = F_v(X)Y + XF_v(Y) - XF_v(I)Y$$

Be careful, this is really saying

$$F_v(XY) = DR_Y F_v(X) + DL_X F_v(Y) - DL_X DR_Y F_v(I)$$

Group Affine System

For example

$$\dot{X} = XU + WX \tag{1}$$

Show that this is group affine!

For constant U and W then the system is integrable

$$X(t) = \exp(tW)X(0)\exp(tU)$$

That is $\Phi_t : \mathbf{G} \rightarrow \mathbf{G}$

$$\Phi_t(X_0) = \exp(tW)X(0)\exp(tU)$$

defines the solution of (1) for any initial condition $X_0 \in \mathbf{G}$.

Error dynamics are observer-independent

Consider a system

$$\dot{X} = F_v(X)$$

that is group-affine.

Consider an observer

$$\dot{\hat{X}} = F_v(\hat{X})$$

Then the error $E = \hat{X}X^{-1}$ dynamics for a group-affine system are observer-independent

$$\dot{E} = F_v(E) - EF_v(I)$$

This is not true for a general equivariant system.

$$\dot{E} = F_{\psi(\hat{X}, v)}(E) - EF_{\psi(\hat{X}, v)}(I)$$

Linear Systems are Group Affine

Euclidean space is a group under addition $\xi + \zeta$

$$\xi^{-1} = -\xi, \quad I = 0.$$

Consider a system on $\mathbb{R}^n$

$$\dot{\xi} = F_u(\xi) = A\xi + Bu$$

Then

$$F_u(\xi + \zeta) = (F_u(\xi) + \zeta) + (\xi + F_u(\zeta)) - (\xi + (F_u(0) + \zeta))$$

prove on board!

Linear Systems are not invariant

A left invariant system $\dot{X} = F_U(X)$ on a Lie group is

$$\dot{X} = XU = DL_X U = DL_X F_U(I)$$

Right invariant is analogous.

For the Euclidean group symmetry one has $L_\xi \zeta = \xi + \zeta = R_\xi \zeta$ and

$$DL_\xi \zeta = \frac{d}{d\zeta}(\xi + \zeta) = I = DR_\xi \zeta.$$

Hence, an invariant system on Euclidean space is given by

$$\dot{\xi} = DL_\xi F_v(0) = A0 + Bu = Bu$$

The only linear system that is invariant is when $A \equiv 0$.

Symmetry of Linear systems

Let $Z \in \text{GL}(n)$, and $x \in \mathbb{R}^n$ and give them dynamics

$$\dot{Z} = -ZA, \quad \dot{x} = -ZBu$$

Then set

$$\xi = Z^{-1}(\xi_0 - x)$$

$$\dot{\xi} = -Z^{-1}(-ZA)Z^{-1}(\xi_0 - x) + Z^{-1}(ZBu) = A\xi + Bu$$

Note that

$$\begin{aligned} \dot{Z} &= -ZA, & \text{Left invariant with body velocity } -A \in \mathfrak{gl}(n) \\ \dot{x} &= -ZBu, & \text{Invariant linear system with body velocity } -ZBu \in \mathbb{R}^n \end{aligned}$$

Symmetry of Linear systems

Let $\mathbf{R}^+(n) = \text{GL}(n) \ltimes \mathbb{R}^n$ be group with multiplication

$$(Z_1, x_1)(Z_2, x_2) = (Z_1 Z_2, Z_1 x_2 + x_1)$$

Define a **right** group action $\phi : \mathbf{R}^+(n) \times \mathbb{R}^n \rightarrow \mathbb{R}^n$ by

$$\phi((Z, x), \xi) := Z^{-1}(\xi - x)$$

Consider the left invariant system on $\mathbf{R}^+(n)$

$$\frac{d}{dt} \begin{pmatrix} Z & x \\ 0 & 1 \end{pmatrix} = \begin{pmatrix} Z & x \\ 0 & 1 \end{pmatrix} \begin{pmatrix} -A & -Bu \\ 0 & 1 \end{pmatrix}$$

Define

$$\xi = \phi((Z, x), \xi_0)$$

Then

$$\dot{\xi} = A\xi + Bu$$

Symmetry of dual invariant system

Consider

$$\dot{P} = PU + WP$$

Let $X, Z \in \mathbf{G}$,

$$\dot{Z} = -ZW$$

$$\dot{X} = X \operatorname{Ad}_Z(U + W)$$

Let $P = Z^{-1}XZ$

$$\begin{aligned}\dot{P} &= \frac{d}{dt} Z^{-1} X^{-1} Z \\ &= WZ^{-1}XZ - Z^{-1}XZW + Z^{-1}X \operatorname{Ad}_Z(U + W)Z \\ &= WP - PW + P(U + W) = WP + PU\end{aligned}$$

Lifted group

Define the lifted group $\mathbf{G}^+ = \mathbf{G} \ltimes \mathbf{G}$

$$\begin{aligned}(Z_1, X_1)(Z_2, X_2) &= (Z_1 Z_2, X_1 Z_1 X_2 Z_1^{-1}) \\ (Z, X)^{-1} &= (Z^{-1}, Z^{-1} X^{-1} Z)\end{aligned}$$

With group action $\phi : \mathbf{G}^+ \times \mathbf{G} \rightarrow \mathbf{G}$

$$\phi((Z, X), P) := Z^{-1} P X Z$$

Consider the left invariant dynamics on the lifted group

$$\frac{d}{dt}(Z, X) = DL_{(Z, X)}(-W, W + U)$$

Define

$$P = \phi((Z, X), P_0)$$

Then

$$\dot{P} = P U + W P$$

Automorphism Group

What is the role of $Z \in GL(n)$ for the linear system?

Define $\sigma_Z : \mathbb{R}^n \rightarrow \mathbb{R}^n$

$$\sigma_Z(\xi) = Z\xi$$

is a **general linear automorphism** of $\mathbb{R}^n$.

For dual invariant systems $Z \in \mathbf{G}$ defines an **inner automorphism**

$\sigma_Z : \mathbf{G} \rightarrow \mathbf{G}$

$$\sigma_Z(X) = ZXZ^{-1}$$

Linear system case

For linear systems the automorphism capture the homogeneous solution of the system

$$\dot{Z} = -ZA$$

what is left is the in-homogeneous part

$$\dot{x} = -ZBu$$

The solution is constructed by adding the two bits together

$$\xi = Z^{-1}(\xi_0 - x)$$

$$\dot{\xi} = AZ(\xi_0 - x) + Bu = A\xi + Bu$$

Dual invariant systems

For dual invariant systems the automorphism captures the part of the solution that is not left invariant

$$\dot{Z} = -ZW$$

what is left is the in-homogeneous part

$$\dot{X} = X \operatorname{Ad}_Z(U + W)$$

Define

$$P = \phi((Z, X), P_0) = Z^{-1}PXZ$$

Then

$$\dot{P} = PU + WP$$

Structure theory for Group Affine systems

Group affine systems can be thought of splitting into two parts

- The first part is associated with driving terms that generate an automorphism of the state space.
- The second part is a left invariant system for which we have established observer design methodologies.

This structure is extremely powerful since it will allow us to define and observer architecture with nice error dynamics.

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Observer architecture

Consider a dual invariant system

$$\dot{X} = XU + WX$$

Propose an internal model for the observer

$$\dot{\hat{X}} = \hat{X}U + W\hat{X}$$

Consider the error

$$E = \hat{X}X^{-1}$$

Then

$$\dot{E} = WE - EW$$

Error automorphism

Note that these error dynamics look like the differential of the automorphism $\sigma_Z(X) = Z^{-1}XZ$ for

$$\dot{Z} = -ZW.$$

That is for E_0 constant define

$$E = Z^{-1}E_0Z$$

then

$$\dot{E} = WE - EW$$

Lifted observer architecture

Consider a dual invariant system

$$\dot{X} = XU + WX$$

Propose an observer

$$\begin{aligned}\dot{\hat{X}} &= \hat{X}U + W\hat{X} \\ \dot{Z} &= WZ\end{aligned}$$

Consider the **lifted error**

$$E^+ = Z^{-1}\hat{X}X^{-1}Z$$

Then

$$\dot{E}^+ = 0$$

Synchronous models

We say that

$$\begin{aligned}\dot{\hat{X}} &= \hat{X}U + W\hat{X} \\ \dot{\hat{Z}} &= W\hat{Z}\end{aligned}$$

contains a **synchronous model** of

$$\dot{X} = XU + WX$$

That is for no correction terms, then

$$\dot{E}^+ = 0$$

Notes on Observer Design

Let $\mathcal{L} : \mathbf{G} \rightarrow \mathbb{R}_+$ be a Lyapunov function. Then

$$\frac{d}{dt}\mathcal{L}(E^+) = 0$$

if the correction terms are set to zero.

If the correction terms in the observer are set to zero then

$$\frac{d}{dt}\mathcal{L}(E^+) = D\mathcal{L}(E^+)\dot{E}^+ = 0$$

Caveat: You must ensure that σ_Z does not become ill conditioned!

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